Lil'wat First Nations Woodland License

Data Package & Base Case Results

Version 2.3

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Project 1282-3

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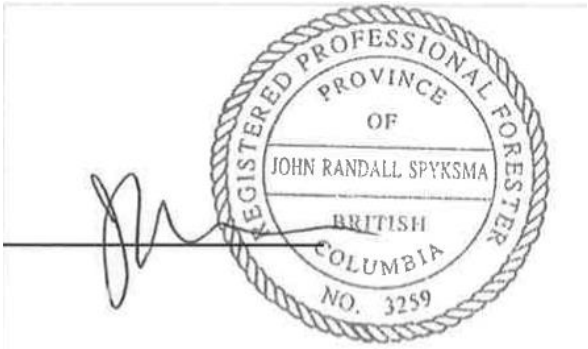
Signatures of Preparing and Supervising Foresters

Kat Gunion, RPF #4915 Forsite Consultants Ltd. Preparing Forester	

 **Randy Spyksma, MSc, RPF #3259** Forsite Consultants Ltd. Preparing Forester	*"I certify that the work described herein fulfills the standards expected of a member of the Association of British Columbia Forest Professionals and that I did personally supervise the work"*

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2 Introduction

Lil'wat Forestry Ventures, wholly owned by the Lil'wat Nation, is currently investigating obtaining a First Nation Woodland License (FNWL). The tenure surrounds the community of Pemberton and the Mount Currie IR. The proposed area encompasses 142,262 hectares within the current boundary of the SOO Timber Supply Area (TSA) and the Sea to Sky Forest District. 63,363 hectares within the boundary is considered productive forest land. Of this, 14,528 hectares is estimated to be available for timber harvesting. Publicly available planning documents are listed in Table 1.

Table 1 Publicly available planning documents

Plan Type	Plan Title	Description	Web Link (as of Dec 2017)
Regional Land use Plan	Sea To Sky Land and Resource Management Plan (2008)	Documenting land use planning within the Sea to Sky corridor.	https://www.for.gov.bc.ca/tasb/slrp/plan79.html
Forest Stewardship Plan (FSP)	PEBBLE CREEK TIMBER LTD. CREEKSIDE RESOURCES INC. LIL'WAT FORESTRY VENTURES LIMITED PARTNERSHIP FOREST STEWARDSHIP PLAN (2012)	This FSP specifies results and Strategies to comply with the Forest Range and Practices Act (FRPA). Forestry activities that fall under this FSP must adhere to the results and strategies outlined in this document.	https://www.for.gov.bc.ca/ftp/DSQ/external/!publish/Forest%20Stewardship%20Plans/Pebble%20Creek%20Creekside%20%20212/Extension-Amendment2%20FSP%203.0.pdf
Timber Supply Review (TSR)	Soo Timber Supply Area Timber Supply Review (Data Package 2008)	This document outlines the assumptions that were made when defining the AAC for the SOO TSA.	https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/forestry/stewardship/forest-analysis-inventory/tsr-annual-allowable-cut/soo_tsa_data_package.pdf
Landscape Unit Objective Orders	BILLYGOAT LANDSCAPE UNIT (2004)	Landscape unit objectives define Wildlife Tree Retention Areas (WTRAs) required for in-block retention based by Bec Zone and Sub zone. Landscape unit objectives also set forth objectives for Old Growth Management Areas (OGMAs).	https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Billygoat_LUP_Order_Objectives_July8_04.pdf
	Birkenhead and Gates Landscape units (2005)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Birkenhead_LU_Legal_Order_Objectives.pdf
	Lizzie, Mamquam,Tuwasus, Solquet High Solquet North, Solquet South Landscape Units (2014)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Mamquam_Lizzie_Tuwasus_Sloquets_OGMA_Order_Revised_October30_2014.pdf
	Meager Landscape Unit (2004)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Meager_LUP_Order_Objectives_July8_04.pdf
	Railroad Landscape Unit (2004)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Railroad_LUP_Order_Objectives_July8_04.pdf
	Ryan Landscape Unit (2004)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Ryan_LUP_Order_Objectives_July8_04.pdf
	Soo Landscape Unit (2004)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Soo_LUP_Order_Objectives_July8_04.pdf
	Amendment to Soo Landscape unit (2010)		https://www.for.gov.bc.ca/tasb/slrp/srmp/coast/sea_to_sky/reports/Pages_from_SooLU_ApprovedBoundaryAmendment.pdf

This document outlines the land base, growth and yield, management and modelling assumptions, and reports the results of the timber supply analysis work completed in support of the development of this new tenure.

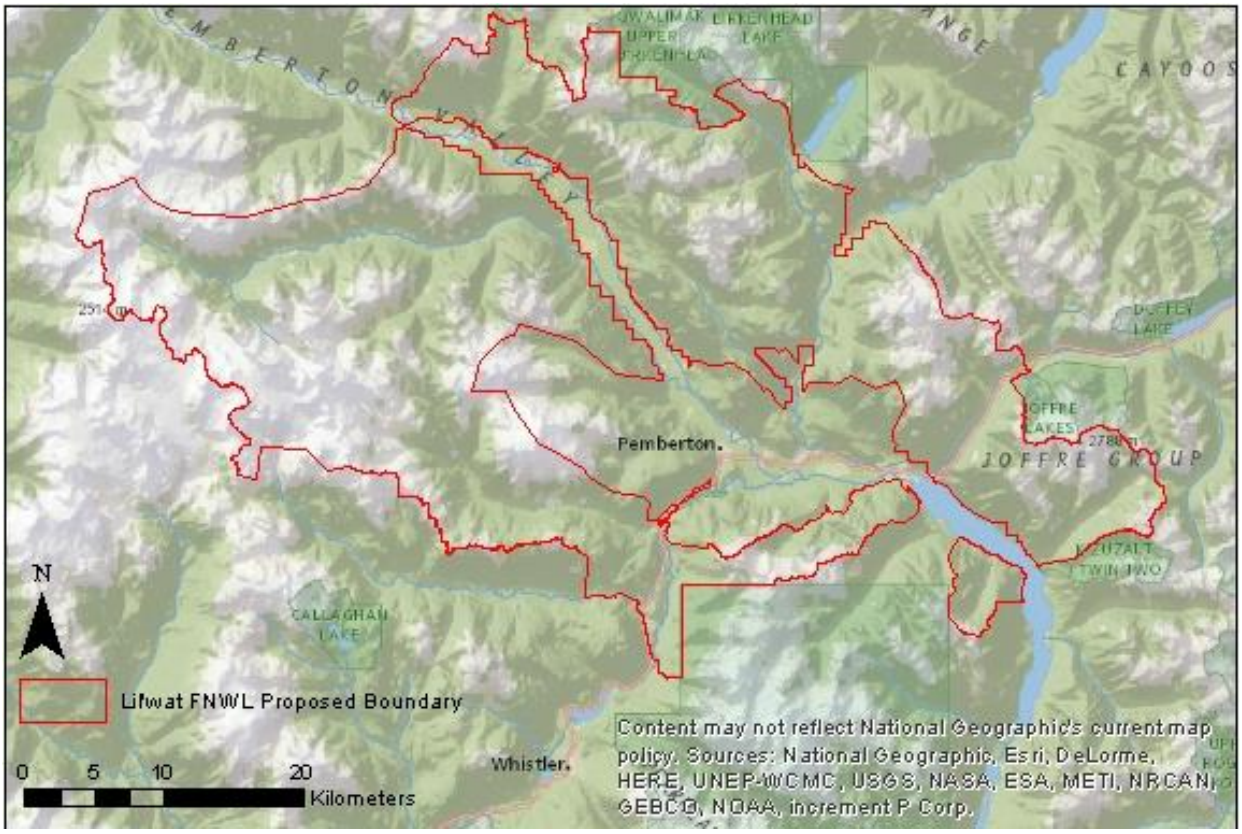

Figure 1 Location Map of Lil'wat FNWL.

3 Land Base Definition

3.1 Inventory

Vegetation Resources Inventory (VRI) was used for this analysis. Inventory attributes (Age) attributed to 2016 were photo captured between 1964 and 2015. FLNRO projected stand attributes to 2016, and Forsite projected ages forward one year to 2017. Additionally, depletions were reflected using Forest Tenures (FTEN) Cut Blocks, Reporting Silviculture Updates and Land Status Tracking System (RESULTS) Forest Cover, RESULTS Forest Reserves, and RESULTS Openings current to October 2017.

3.2 Data Sources

Several datasets covering administrative, inventory, and management guidance information were included in the analysis (Table 2). These datasets were processed to develop a resultant dataset that was used to build the forest estate model.

Table 2 Data sources used (some may not be pertinent to this project, but were all compiled for modelling).

Data Class	Data Description	Source	Forsite Internal Name	Vintage
Base	Utility Corridors, Gas, Hydro	LRDW	TANTALIS_CROWN_RIGHTS_OF_WAY	2017
Tenure	Forest Ownership	static	FOREST_VEGETATION_F_OWN	2017
	remaining Timber Licences	FLNRO	Remaining_TL	
	Consolidated Tenure layer including, CFA, FNWL, WL1666, IR6&7	Forsite	UNION_TENURES	
Forest	Biogeoclimatic Ecosystem Classification (BECv10)	LRDW	BEC	2016
	Forest Cover with depletions from Results, FTA. Reserves, and managed site productivity.	LRDW	VRI_SI_DEP	2017
	WL 1666 Permanent WTRAs	Chartwell	W1666_Reserve	2017
LUP	Landscape Units	LRDW	LU	2017
	OGMAS	LRDW	LAND_USE_OGMAS	2016
	Category A Spirited Places. Selected from LAND_USE_PLANNING_RMP	LRDW	Spirited_Category_A	2017
	Category B Spirited Places Selected from LAND_USE_PLANNING_RMP	LRDW/forsite	Spirited_Category_B	2017
	Wildand_Zones Selected from LAND_USE_PLANNING_RMP	LRDW/forsite	Wild_Land_Zones	2017
	Floodplain management Areas Selected from LAND_USE_PLANNING_RMP	LRDW/forsite	Floodplain	2017
	Upper Birkenhead Corridor (only old) Selected from LAND_USE_PLANNING_RMP	LRDW/forsite	Upper_Birken_Minus_Young_Stands	2017
	Lillooet lake RMZ – a 100m buffer around Lillooet Lake	LRDW/forsite	Lillooet_lake_RMZ	2017
Resource	Community Watershed	LRDW	WLS_COMMUNITY_WATERSHED	
	Operability – areas classified as: inoperable, helicopter, conventional	FLNRO	OPERABILITY	2011
	Visual Landscape Inventory	LRDW	FOREST_VEGETATION_REC_VISUAL	2017
	Terrain Stability Assessement	LRDW	STABILITY_CLASS	2017
	Pemberton Creek Buffer – Pemberton creek buffered by 100m	Forsite	Pemberton_creek_Buffer	2017
	ESAs	MFLNRO	ESA	2011
	Riparian Buffers – Created from streams, river poly, lakes	Forsite	Riparian_buffered	2017
	Road Buffer - based on Road Segments and Road Atlas, buffered	Forsite	Road_Buffer_All	2017
Wildlife	UWR	LRDW	WILDLIFE_MANAGEMENT_UWR_for_resultant	2017

	Wildlife Habitat Areas (WHA)	LRDW	WILDLIFE_HABITAT_AREAS	2017

Land base assumptions are used to define the contributing forest landbase (CFLB) and the timber harvesting land base (THLB) in the study area. The THLB is designated to support timber harvesting while the CFLB is identified as the broader productive forest that can contribute toward meeting non-timber objectives (e.g. biodiversity). The CFLB includes the THLB, and the area within the CFLB that is not harvestable, is called the non-harvesting landbase (NHLB). In the FNWL, the effective long term THLB is 14,757ha (10.4% of the gross area).

Table 3 Area Land Base Assignments

Land Base element	Total Area (ha)	Effective Area (ha)*	% Total Area	% CFLB
Total Crown area (NW TSL)	**142,262**	**142,262**	**100%**	
Less:				
Non/Veg Non Forested	79,899	78,038	54.9%	
Utility Right of Ways	461	191	0.1%	
Existing Roads	815	670	0.5%	
Contributing Forest Landbase (CFLB)		**63,363**	**45%**	**100%**
Less:				
Spirited Grounds A	862	715	0.5%	1.1%
Spirited Grounds B	608	367	0.3%	0.6%
Cultural	1,296	875	0.6%	1.4%
Wildland	72,492	12,287	8.6%	19.4%
OGMA	6,394	4,097	2.9%	6.5%
WHA Grizzly	6,774	917	0.6%	1.4%
WHA LTOH	9,747	6,577	4.6%	10.4%
UWR	14,462	2,778	2.0%	4.4%
Crown Reserves	14,474	335	0.2%	0.5%
Steep Slopes	106,910	13,985	9.8%	22.1%
ESA	16,442	471	0.3%	0.7%
Low Timber Growing	61,346	3,278	2.3%	5.2%
Pine Low SI	1,909	34	0.0%	0.1%
Unmerch Species	1,374	0	0.0%	0.0%
Riparian	3,070	348	0.2%	0.5%
Isolated	5	5	0.0%	0.0%
Temporary Exclusions:				
Timber License	371	183	0.1%	0.3%
Reserves	350	153	0.1%	0.2%
Spatial Timber harvesting land base - Short Term		**15,958**	**11.2%**	**25.2%**
Plus:				
Timber License		183		
Reserves		153		
Spatial Timber harvesting land base - Long Term		**16,293**	**11.5%**	**25.7%**
Riparian (streams) 2%		274	0.2%	0.4%
Stand Level Retention- WTRA		1,102	0.8%	1.7%
Spotted Owl MFHA		390	0.3%	0.6%
Effective timber harvesting land base		**14,528**	**10.2%**	**22.9%**

Less Future Non-Spatial Netdowns:			
Future permanent roads (0.6%)	92	0.1%	0.1%
Effective future timber harvesting land base	**14,435**	**10.1%**	**22.8%**

*Effective netdown area represents the area that was actually removed as a result of a given factor. Removals are applied in the order shown above. Areas removed lower on the list do not contain areas that overlap with factors that occur higher on the list. For example, lake buffers netdown does not include non-forested area.

**Aspatial netdowns are applied in the model or yield curves and are not reflected in the GIS dataset areas.

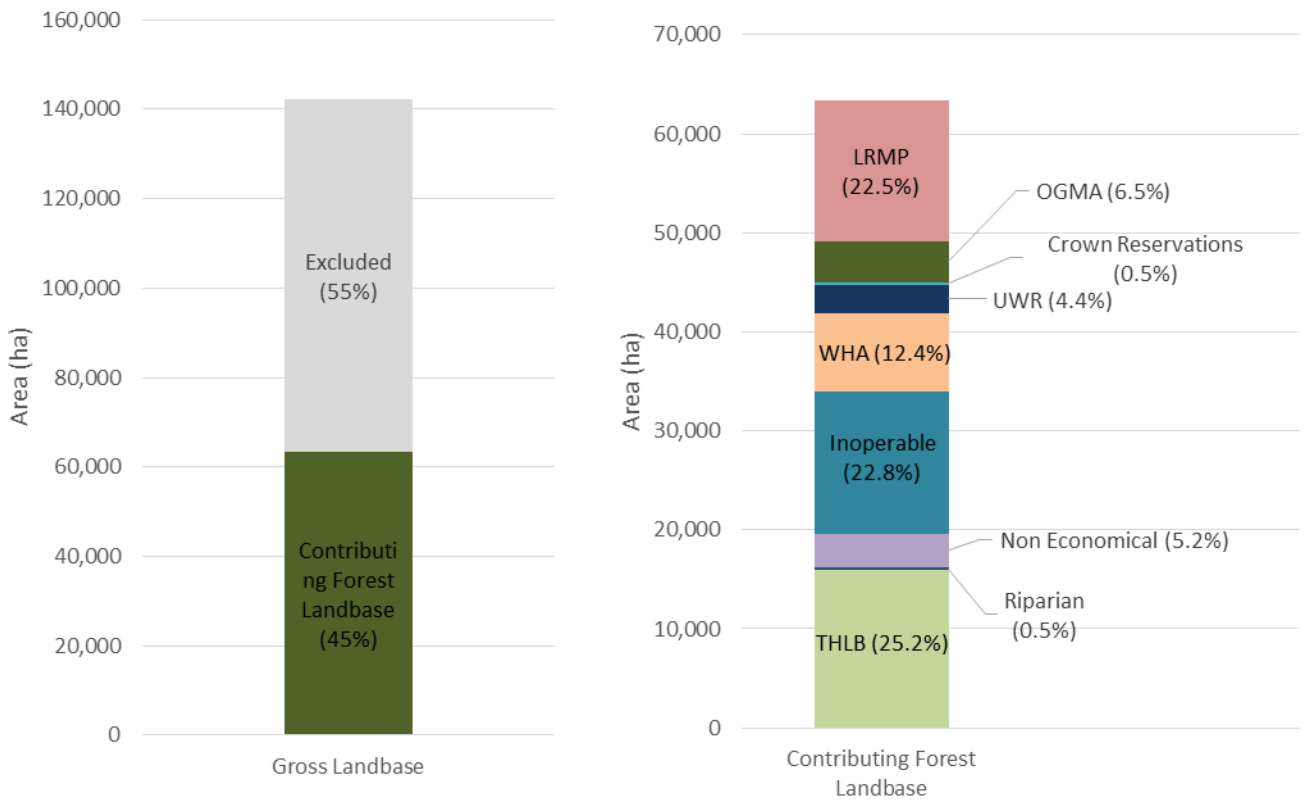

Figure 2 Graphical Representation of Landbase Breakdown

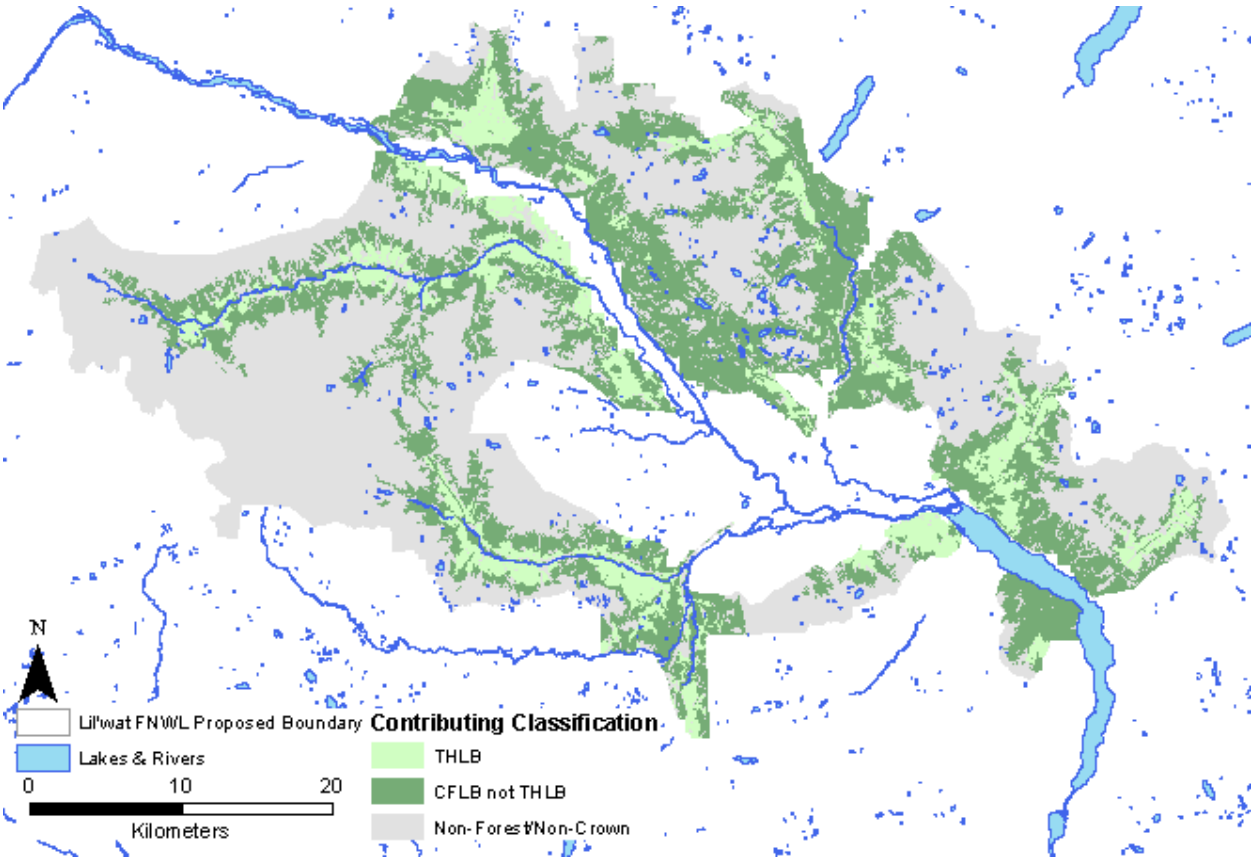

Figure 3 Landbase classification map.

3.3 Exclusions from contributing forest

Exclusions are removed from the analysis and do not contribute to timber harvesting or non-timber constraints.

3.3.1 Non-Vegetated / Non-Forested

The BCLCS land classifications system was used to define Non-vegetated and non-forested areas. Forest is defined as polygons classified as treed vegetation on wetlands or uplands with a site index of 5 or greater. In addition, any polygons with a previous harvesting history are considered to be forested regardless of the BCLCS classification.

3.3.2 Utility Right of Ways

Right-of ways for Transmission lines, electric power lines, and water power lines are all spatially removed from the landbase. These rights of way were identified using the Tantalus data, and cross referenced with spot imagery. There are 461 hectares of right of ways in the FNWL, of which 191 hectares are effectively removed from the landbase.

3.3.3 Existing Roads

Existing roads come from the BC road atlas and FTEN Road Segments. They are buffered based on road class and road description. Main roads are buffered by 26 metres (13 metres each side of a linear feature, and smaller operational roads are buffered to a 10 metre width (5m each side of the linear

feature). There are 814 hectares of buffered roads within the FNWL, of which 669 hectares are effectively removed from the landbase.

3.4 Reductions from THLB

Forested lands that are removed from the THLB can still contribute to non-timber objectives (e.g., VQO, Green-up, Wildlife habitat). This section describes the gross area that each objective constrains, as well as the effective THLB removal. The effective THLB removal is the portion of the gross area that has not been removed already for other reasons (e.g., non-forest, road etc.).

3.4.1 LRMP Objectives

The Sea to Sky LRMP was adopted in 2008. Within the LRMP there are spatial reductions for Cultural Places, Cultural Management Areas, and Wildand zones. These areas are described in this sub section. Non-timber management associated with the LRMP is discussed in sub-section 4.1.

Cultural Places

Lil'wat, Category A Spirited Grounds, one Squamish Nation Cultural Site, and a subset of category B Spirited Grounds were removed from the THLB. Table 4 and Table 5 show the gross areas of each of the cultural places within the FNWL.

Table 4 Cultural Places to be removed from THLB.

Cultural Places	Area (ha)
6 Mile Creek spirited ground area	1
Hindu Flats spirited ground area	47
The Lil'wat Nation Q'em'sam Old Growth and Sensitive Area	500
Lower Birkenhead spirited ground area	62
Lower Soo River Síiyamín ta Skwxwú7mesh (Cultural Site)	107
North Millar spirited ground area	15
Owl Creek Pool spirited ground area	23
Shadow Lake spirited ground area	3
Spetch Creek spirited ground area	32
Upper Birkenhead spirited ground area	64
Wolverine Lake spirited ground area	10
Total	862

In total 1470 hectares are identified within the FNWL because they are cultural places (A or B), overall, Cultural places are responsible for 1082 hectares being removed from the THLB.

Table 5 Category B Spirited Ground areas to be removed from THLB

Spirited Area	Area (ha)
Sampson Creek spirited ground area	608

Cultural Management Areas

The Birkenhead corridor is part of a Cultural Management area and was removed from the THLB with the exception of lands which were less than 40 years old in 2008. These polygons less than 40 years old in 2008 have been delineated in the LUP spatial Layer and were not included in the Birkenhead corridor layer. The gross area of the cultural removals are 1,295 ha, with 874 hectares of effective removal area.

Wildland Zones

The wildland zones are defined in the LRMP and they are to be removed from harvest. Table 6 describes the zones that overlap the FNWL.

Table 6 Wildland Zones for Cultural, Recreation, Tourism, and Wildlife.

Type	Name	Area
Cultural	Qwalímak / Birkenhead Headwaters	12,532
	Skelulátkwa / Owl Creek	4,270
	Soo Headwaters	4,189
	Upper Elaho	64
	Úll'us / Ryan Headwaters	18,764
Recreation	Barr Headwaters	1,086
	Mount Currie	3,379
	Pebble Creek	560
	Place Glacier Ridge	4,307
	Rutherford Headwaters	13,294
	South Creek	255
	Twin One Headwaters	6,087
Tourism	Birkenhead Peak	3
Wildlife	Sockeye Headwaters	10
	Wolverine Creek	3,690
Total		72,492

3.4.2 Old Growth Management Areas (OGMAs)

All of the landscape units within the FNWL have approved OGMAs. These will be used to satisfy old seral requirements and were removed from the THLB. There are 6,394 hectares of approved OGMAS across seven landscape units. 4,097 hectares are effectively removed due to OGMAS.

Table 7 OGMA gross area by LU.

LU	Area (ha)
Billygoat	324
Birkenhead	1,913
Lizzie	691
Meager	86
Railroad	1,286
Ryan	1,460
Soo	635

Total	6,394

3.4.3 Wildlife Habitat Areas (WHAs)

Designated WHAs for Long Term Owl Habitat (LTOH) and grizzly bear habitat have been delineated and were removed from the THLB. LTOH covers a gross area of 9,747 hectares while effectively removing 6,577 ha, and the Grizzly habitat covers 6,774 hectares and removed 917 from the THLB.

Areas within the Spotted Owl MFHA have a requirement of leaving 40 stems per hectare of the largest 80 stems per hectare. This will be emulated in the model by implementing a constraint to reserve an additional 20% aspatial retention for blocks within these polygons. Within the FNWL this will effectively remove 390 hectares.

3.4.4 Ungulate Winter Ranges (UWRs)

Two UWR orders cover the area, including u-2-002 for Goat Winter Range, and u-200-5 for Moose Core Winter Range and Retention Deer Winter Range. All of these areas were removed from the THLB effectively removing 2,778 ha. Total netdown classification of the no harvest UWRs are listed in Table 8. Note that black tailed deer and mountain goat overlap.

Table 8 Resulting netdown classification for No Harvest UWRs.

Mountain Goat (#U-002)	Moose (#U2-005)	Black Tailed Deer (#U-005)	NHLB	Excluded
		NO HARVEST ZONE	1,961	299
	NO HARVEST ZONE		64	10
NO HARVEST ZONE			4,787	7,141
NO HARVEST ZONE		NO HARVEST ZONE	150	51
Total			**6,961**	**7,501**

This section describes the reductions from the THLB, however U-005 also has Conditional Harvest Zones which are discussed in section 4.2.

3.4.5 Parks and Conservancies

The boundaries for the FNWL have been carefully delineated around any parks and conservancies in the area. No further reductions were required.

3.4.6 Land Ownership Removals

Polygons from the Forestry Ownership Layer categorized as 68U (Biodiversity, Mining, and Tourism, and 69U (Crown Miscellaneous Reserves) were evaluated to be removed from the THLB. In total 335 hectares are removed for ownership.

3.4.7 Operability Restrictions

Steep Slopes

FLNRO developed an operability layer for TSRII and improved it for TSR3. This layer distinguishes between conventional logging, (cable, grapple, yarder), helicopter logging, and inoperable areas. Inoperable areas were removed from the THLB except for areas where previous harvest has occurred. This resulted in 13,985 hectares being removed from the THLB due to inoperable conditions. A total of

464 hectares (3% of what was removed) were identified as inoperable, but remained in the THLB because they had been previously harvested.

The helicopter and conventional classifications are used when assigning growing potential based on minimum volume requirements (section 3.4.8).

Table 9 Operability classifications within the FNWL.

Operability	Area (ha)
Conventional	25,300
Helicopter	7,222
Inoperable	**106,826**

Unstable Soils /Environmentally Sensitive Areas

Terrain stability mapping has only been completed within the Pemberton Community Watershed. The class U and V polygons are outside the FNWL.

In the remainder of the area, the Environmentally Sensitive area mapping from FLNRO was used. Any polygons that have been labelled as ESA_1 (identified sensitive) for soils, snow, regeneration, or recreation were fully removed from the THLB. The gross area classified as ESA is 16,442 ha with 471 ha removed for this reason.

Table 10 Environmentally sensitive classifications

ESA Category	Area (ha)
Snow	111
Regeneration	4,480
Recreation	324
Soils	11,527
Total	16,442

3.4.8 Low Timber Growing Potential

Stands with low timber growing potential were removed from the THLB based on achieving a minimum volume at a cut off age, depending on harvesting method and species as outlined in Table 11. If the stand is currently older than the cut off age and exceeds the required volume, then it is still considered merchantable. In addition to these requirements, pine stands must have a managed site index of greater than or equal to 20 metres.

Table 11 Minimum Requirements for Growing Potential.

Leading Species	Age Cut off	Projected/Current Volume Conventional	Projected Volume Requirements Heli
Fir	140	350 m^3/ha	400 m^3/ha
Cedar	140	350 m^3/ha	400 m^3/ha
Hemlock/Balsam	140	350 m^3/ha	400 m^3/ha
Spruce	140	300 m^3/ha	400 m^3/ha
Pine	140	300 m^3/ha	350 m^3/ha

3.4.9 Non-merchantable forest types

Ponderosa pine, larch, alder, birch, cotton wood and maple were removed from the THLB. Although 1,374 hectares of these non-merchantable forest types were identified, all of these types were removed for other netdown reasons resulting in an effective netdown area of zero.

3.4.10 Riparian

Freshwater Atlas Rivers, Lakes, and Wetlands were obtained from DataBC. The large rivers were identified and classified from field experience. The riparian reserve zone width and the basal area retention level applied to each riparian stream class (Table 12).), meet or exceed the requirements for retention within the riparian management zone, as defined by the Forest Practices and Planning Regulation (FPPR). For large streams (S1, S2 and S3 features) the area buffered was calculated by summing the riparian reserve zone and 20% of the riparian management zone. All other streams were then buffered by 5m each side. The proportion of the small streams that cover the THLB was then applied as an aspatial reduction of 2% from each THLB polygon.

Lakes and wetlands were classified by size and a buffer applied spatially on the landbase. Lillooet Lake has a special buffer discussed in section 4.1.

Table 12 Riparian Buffers Applied

Riparian Class	Size	Description	Buffer Width(m)			
			Riparian Management Area (m)	Riparian Reserve Zone (m)	Riparian Management Zone (m)	Effective Riparian Buffer (m)
Streams	S1 – B	Fish Bearing & > 20m Wide	70	50	20	54
	S2	Fish Bearing & 5m to 20m wide	50	30	20	34
	S3	Fish Bearing & 1.5m to 5m wide	40	20	20	44
	Small	All other Fresh water atlas	10 m(5m each side) Applied aspatially in modelling			
Lakes and Wetlands	Large	Area ≥ 5ha				20 m
	Medium	Area ≥1ha <5ha				15 m
	Small	Area <1ha				20m

3.4.11 WTRAs and in-block Retention

Seven landscape units have legal objectives defined for the amount of wildlife tree retention required (Table 13). The remaining area (which includes the Lizzie and Upper Squamish LUs), will have the FPPR default of 7% applied. These will be modelled as an aspatial THLB reduction. In patchworks, this percentage is assigned to blocks as an 'unmanaged' portion which does not get harvested ever in the model, and is able to contribute to old targets.

Table 13 Landscape Unit/BEC sub zone combinations covered by a legal order for WTRA percents.

Landscape Unit	Bec sub zone	Percent WTRA
Billygoat	IDF ww	12
	CWH ds	7
	CWH ms	2

	MH mm	0
Birkenhead	IDf ww	5
	CWH ds	4
	CWH ms	3
	MH mm	2
	ESSF mw	0
Gates	IDF mm	8
	CWH ds	6
	CWH ms	6
	ESSF mw	1
	MH mm	0
Meager	CWH ds	9
	CWH ms	7
	MH mm	0
Railroad	CWH ds	8
	IDf ww	8
	ESSF mw	0
	MH mm	0
	CWH ms	6
Ryan	CWH ds	10
	CWH ms	8
	MH mm	0
Soo	CWH ds	8
	CWH ms	7
	MH mm	3

The short term THLB does not include current mapped WTRAs which are non-harvestable for a term of 50 years, and Timber Licences that return to the FNWL once their current holder harvests them. These areas are shown in Table 3 being added to the short term landbase.

3.4.12 Isolated stands

After all the THLB was identified, stands that are less than half a hectare, and further than 100m from other THLB or a current road were removed as an isolated stand.

3.4.13 Future Roads, Trails and Landings

In this analysis, the future road reduction is determined to be 2% of the area harvested that is further than 300m from an existing road, using the equation below:

$$\%\ of\ THLB\ as\ Future\ Road = \frac{(\text{Area of THLB} > 300\text{m from road}) * 0.02}{Area\ of\ THLB}$$

$$\frac{(5{,}915) * 0.02}{18{,}590} = (0.6\%)$$

The 2% threshold comes from discussions with ministry where the agreed percent alteration due to roads.

This percentage is applied in the model as a yield reduction for all future managed stands following harvest of existing natural stands. Patchworks does not allow polygon areas to change after model initiation, so all yields are factored for future analysis units.

3.5 Landbase Characteristics

The proposed tenure covers six biogeoclimatic (BEC) zones, including the MH, CWH, ESSF, IDF, CMA, and IMA (Figure 4). CMA and IMA are alpine zones with no productive timber, but polygons within these zones with a site index greater than five are assumed to be forested and are capable of contributing to non-timber forest requirements. The area also covers four natural disturbance types (NDT). These zones and disturbance types drive natural disturbance and productivity on the landbase.

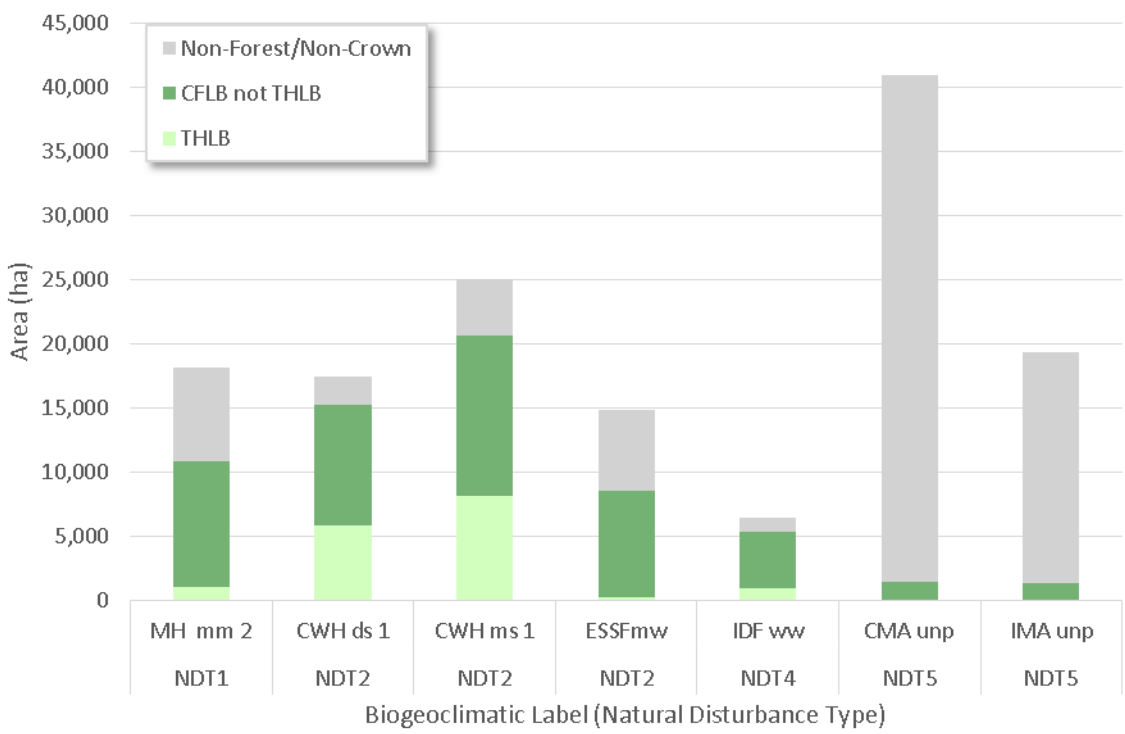

Figure 4 BEC sub zone and NDT distribution on the FNWL

The forested area of the FNWL is mostly covered with Douglas-fir and Amabilis Fir leading stands. Additionally there is cedar, hemlock, pine, deciduous, and spruce leading stands (Figure 5). Any deciduous leading stands that are present within the THLB (60 hectares), are previously harvested stands who are assigned yield curves based on the BEC weighted average.

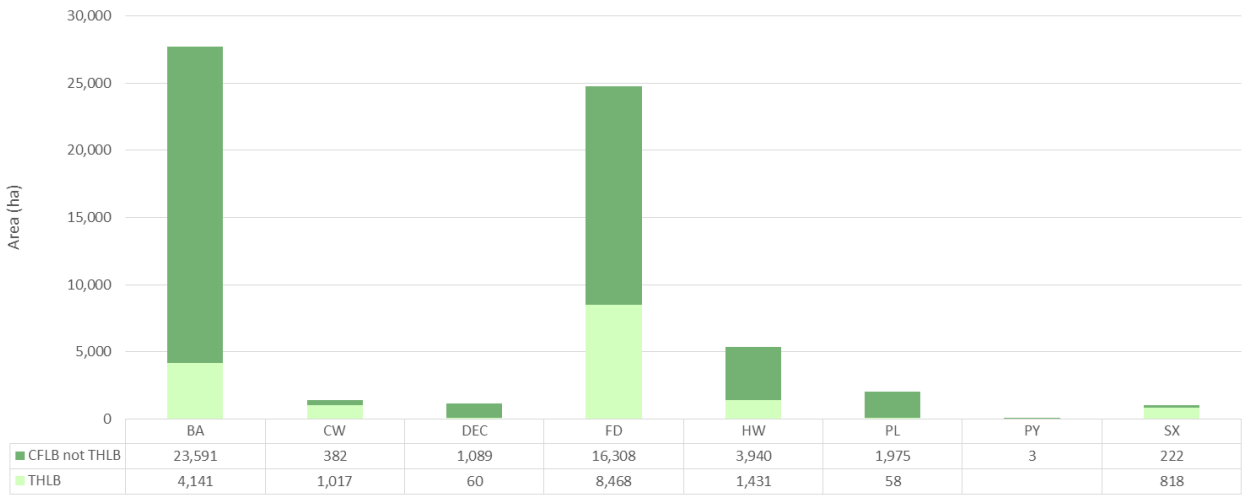

Figure 5 Area by leading species on the CFLB.

The age class distribution of the THLB on the landbase indicates a very young land base (Figure 6). Approximately 11,000 hectares are less than 50 years old. However, much of the NHLB (CFLB not THLB) is older than 200 years.

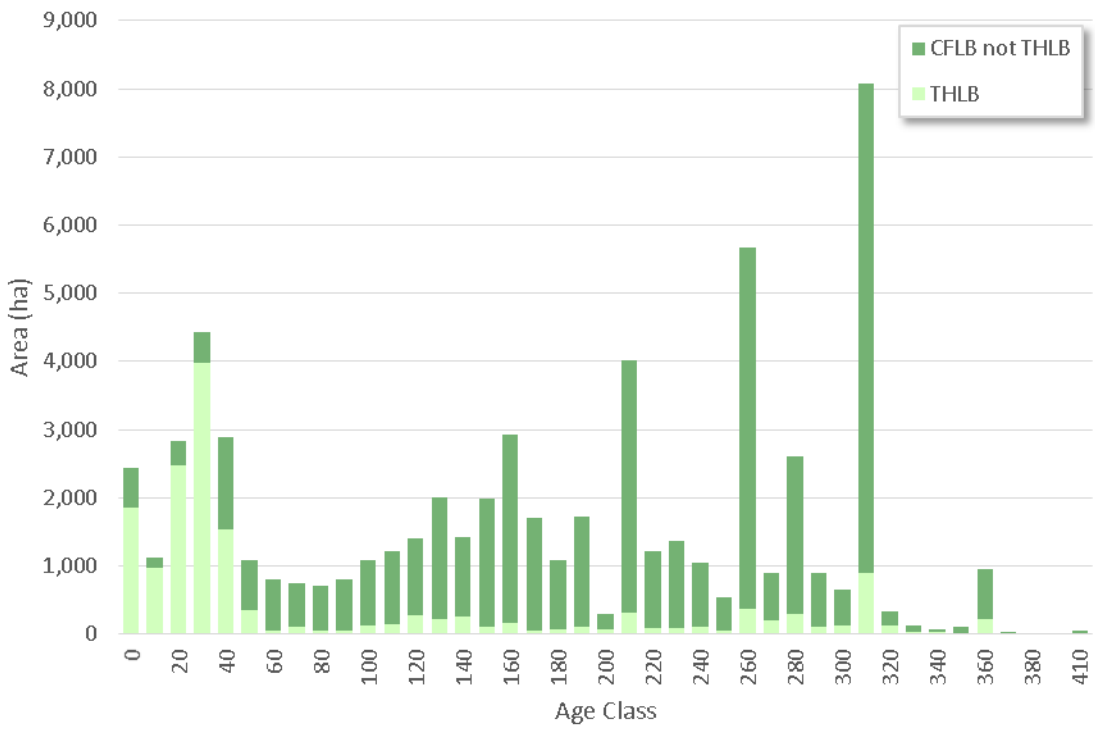

Figure 6 Ten year age class distribution on the CFLB

Site index (SI) is used to define the productivity of a stand. Site index is the height that the best trees on the site will be at 50 years breast height age. There are two SIs, inventory (Figure 7) and managed (Figure 8). Inventory SI comes from the VRI inventory. Managed site index was taken from the SOO TSR resultant. The TSR did a site index adjustment based on work done by timberline in 2008. Instead of replicating this work, each of our resultant polygons was overlayed with the TSR resultant polygons. A

weighted average for each THLB polygon was calculated and then assigned as the managed site index. Stands that did not get a site index from the TSR resultant were assigned a site index based on the Provincial Productivity Layer. The resulting weighted average site index was 22.9 m.

Managed site index is often higher than the inventory site index. On the FNWL, the weighted average is 4.4 metres taller (18.5m inventory SI versus 22.9m managed SI). Site index is discussed further in section 6.5.

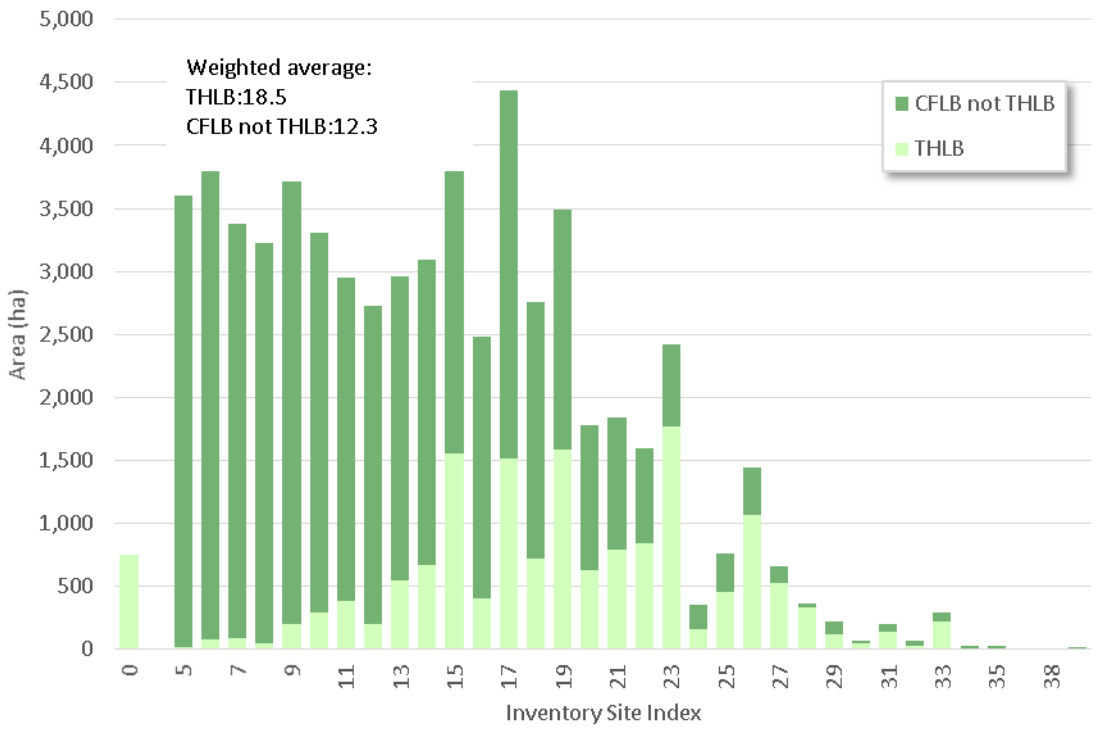

Figure 7 Inventory Site Index

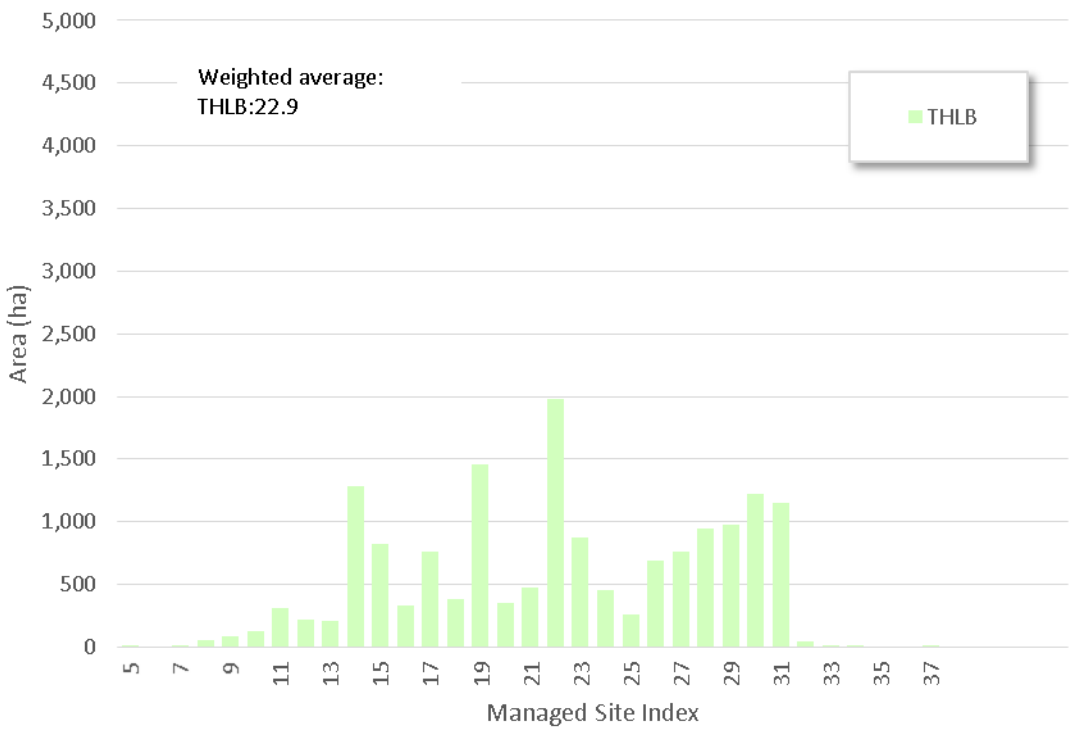

Figure 8 Managed Site index on the THLB from TSR3 site index adjustment.

4 Non-Timber Management Assumptions

Non-timber management assumptions are targets on the landbase that require a certain percentage to meet a certain criteria. An example might be 10% must be in an old seral state, or no greater than 5% may be less than five metres tall. Often these constraints are applied at a spatial level using BEC Zone or LU, stating that these constraints must be applied within each LU, or within a certain watershed.

4.1 LRMP Objectives

Floodplain Management Areas/ Riparian Management

The constraints described in Table 14 are constraints for the LRMP objectives for riparian and floodplain management areas, Within the the floodplain management areas less than 20% will be less than 60 years old and for riparian buffers,greater than 70% must be in a mature or old seral state.

Table 14 Floodplain and Riparian management for the LRMP

Location	Management
Upper Lillooet River Floodplain	➢ 70% Mature/Old [a] ➢ <20% Gross area will be <60 yrs (94.9 ha)
Lillooet Lake 100 m Buffer	➢ 70% Mature/Old
Lower Green River Floodplain	➢ <20% Gross area will be <60 yrs (3 ha)

[a] This target is for the upper Lillooet River Riparian Area and spirited areas adjacent to it.

4.2 Ungulate Winter Ranges (UWRs)

UWR order u-2-005 has two management constraints. First, Deer Rotation Winter Range has two requirements defined as allowing a maximum of 20% of area to be less than 20 years in age, and a minimum of 20% of the area must be older than 100 years. Second, a Moose Forage Management Zone has a constraint at an operational scale and will not be modelled. Table 15 shows the resulting netdowns for the two conditional harvest areas.

Table 15 Resulting netdown classification for UWRs with conditional harvest allowance.

Conditional Harvest Zone	THLB	NHLB	Excluded
Black Tailed Deer (#U-005)	305	495	65
Moose (#U2-005)	3	180	232
Total	**308**	**675**	**297**

4.3 Community Watersheds

There are four community watersheds within the FNWL (Table 16). They were treated the same as they were in TSR, and were modelled by ensuring that no more than 5% of the forested area can be less than five metres tall.

Table 16 Community watersheds

Community Watershed	CFLB (ha)
Catiline Creek	165
McCullock Creek	249
Peq Creek*	0
Rogowski Creek	10
Total	**414**

* All the area within the FNWL from Peq Creek is Non-Forested

4.4 Visual Quality Objectives

VQO Polygons within the FNWL area will were modelled by using Plan to Perspective (P2P) ratios and Visually Effective Green-up (VEG) heights for 5% slope class increments to determine the maximum percent alterations.

The first row of Table 17 splits the landscape into 5% slope classes. These slope classes are used to assign ratios and heights. The second row describes the P2P ratios. These ratios help to describe how the area appears if looking at it from a vantage point (perspective) versus looking at it from the sky (plan). Based on research, on level areas, the amount of denudation observed in plan view is 4.68 times that observed in perspective view. The third column describes how tall a stand must be to look 'non-disturbed', this height increases as the slope increases.

Table 17 P2P ratio and Height by slope class

	Modified Visual Unit Slope Classes for P2P Ratios and VEG Heights														
Slope %	0-5	5.1-10	10.1-15	15.1 -20	20.1 -25	25.1 -30	30.1 -35	35.1 -40	40.1 -45	45.1 -50	50.1 -55	55.1 -60	60.1 -65	65.1 -70	70+
P2P Ratio	4.68	4.23	3.77	3.41	3.04	2.75	2.45	2.22	1.98	1.79	1.6	1.45	1.29	1.17	1.04
VEG Height (m)	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	6.5	7.0	7.5	8.0	8.5	8.5	8.5

Table 18 describes the permissible alteration by VQO polygon type. For this landscape we have chosen the mid-point of the permissible range as our target.

Table 18 VQO by Percent Alterations

VQO	Permissible % Alteration in Perspective View	Proposed % Alteration in Perspective View
Preservation	0	0
Retention	0 – 1.5	0.8
Partial Retention	1.6 – 7.0	4.3
Modification	7.1 – 18.0	12.6
Maximum Modification	18.1 – 30.0	24.1

The P2P ratios and VEG heights assessed by slope class, as well as the allowable VQO percent alterations are detailed in Table 17 and Table 18, respectively. Raster cells (20x20m) were assigned slope classes, and the P2P ratio was calculated by weighting the number of raster cells (i.e. slope class) per VQO polygon in the CFLB.

$$\text{Eg: P2P} = \frac{(\#\text{cells in } 0{-}5)*4.68 + (\#\text{cells in } 5{-}10)*4.23 + \ldots}{total\ \#\ of\ Cells}$$

The percent denudation applied to each Visual Landscape Inventory polygon in the model is calculated as the weighted P2P ratio by slope class multiplied by the proposed percent alteration in perspective view for the polygon. The resulting percent denudation value is then applied as a constraint on the maximum proportion of the polygon that can be below the VEG height at any given time.

The next step is to determine age when the VEG height is achieved for each of the Visual Landscape Inventory polygons. For this purpose, the area weighted site index (i.e., top height in m at age 50) and most prevalent species within each Visual Landscape Inventory polygon are determined based on the proportion of predicted future stands. Once the existing stand is harvested and regenerated, it is possible to estimate the age at which the regenerated stand reaches the previously-determined VEG height. The area-weighted average site index is determined from the provincial site productivity layer, while the prevalent species is determined as the leading species with the highest area in each Visual Landscape Inventory polygon. Then, the site index and species are entered in Site Tools (v.4.1 beta) to determine the age where VEG height is reached. Note that deciduous stands are all considered to be dominated by trembling aspen and Visual Landscape Inventory polygons with no site index estimates are assumed to have the area-weighted site index average for the entire land-base.

The calculations for all of the VQO polygons with THLB are listed in Appendix I. To address visuals across the landbase there is a further target set of patches of trees less than 3m not exceeding 40 hectares.

4.5 Prescribed Wildlife Tree Patches & Stand-Level Biodiversity

Existing reserves have been identified using the RESULTS reserve layer. Each wildlife tree patch is to be locked from harvest eligibility for 59 years from the WTP established date. This to emulate that these WTPs must be locked in for a full rotation. Future WTPs are managed as described in section 3.4.11.

4.6 Timber licenses

Portions of Timber licenses that have not reverted back to the crown will initially be unavailable for harvest. Once the model harvests them (the harvested volume will not contribute to the yield), they will then be available for harvest within the Lil'wat tenure for the following rotations. The harvest of the TL stands will be targeted to be completed in the first decade.

5 Harvesting Assumptions

This section describes the criteria and considerations used to model timber harvesting activities.

5.1 Utilization

Utilization standards used to determine merchantable volumes for this analysis are shown in Table 19.

Table 19 Utilization Standards

Leading Species	Minimum Diameter at Breast Height (DBH) cm	Maximum Stump Height (cm)	Minimum Top Diameter (cm)
Pine	12.5	30.0	10.0
Other ≤ 120 yrs	12.5	30.0	10.0
Other > 120 yrs	17.5	30.0	10.0

Non-merchantable species are also removed from the yield curves. This includes all deciduous, ponderosa pine, larch and yew volume.

5.2 Minimum Harvest Criteria

In order for a stand to be considered economic and eligible for harvest within the model it must meet the minimum volume per hectare (MVH) criteria based on leading species and Harvest Method.

Table 20 Minimum harvestable volume criteria for existing and managed stands

Leading Species	Conventional Landbase	Helicopter landbase
Pine	300 m^3/ha	350 m^3/ha
Cedar/Spruce	300 m^3/ha	400 m^3/ha
All Other	350 m^3/ha	400 m^3/ha

5.3 Silvicultural Systems

The dominant silviculture system used in the tenure is clearcut with reserves (WTPs, riparian reserves, etc.) and this is how treatments will be recognized in the model. This is the silviculture system that the SOO TSA TSR applied.

6 Growth and Yield Assumptions

6.1 Analysis Unit Characteristics

Often natural stands are stratified into analysis units to produce weighted average yield curves for modelling. However in this analysis a yield curve was generated for each individual VRI polygon.

In contrast, the yield curves (both existing and future) for managed stands were generated by stratifying the VRI into analysis units based on leading species and managed site productivity class. Table 21 shows a summary of the analysis units for future managed yield curves.

Table 21 Analysis Unit Stratification Characteristics

AU Name	Leading Species	Site index Range (managed Site Index)
1.F-VG	Douglas-Fir	≥30
2.F-G	Douglas Fir	26-29.9
3.F-M	Douglas Fir	20.0 to 25.9
4.F-P	Douglas Fir	<20
5.CS-GM	Cedar/Spruce	≥ 20
6.CS-P	Cedar/Spruce	<20
7.HB-G	Hemlock/Balsam	≥25
8.HB-M	Hemlock/Balsam	20.0 to 24.9
9.HB-P	Hemlock/Balsam	<20
10. P-A	Pine	≥20

6.2 Stand Projection Models

Yield curves developed for existing natural stands were prepared using the Variable Density Yield Prediction (VDYP) 7 for each forest cover polygon. Existing and future managed stand yield curves for each AU were generated using the Table Interpolation Program for Stand Yields (TIPSY) 4.3. All non-merchantable species (Ponderosa pine, larch, alder, birch, cotton wood and maple) were removed from the yields.

6.3 Decay, Waste, and Breakage

For natural stands, reductions to stand volume for decay, waste and breakage factors were set to the default provincial stand loss factors. These factors were applied in the development of the VDYP7 yield curves.

For managed stands, operational adjustment factors (OAF) are utilized in the TIPSY model. An OAF1 of 15% was applied, while OAF2 increases from 0% to 5% by the time the stands reach 100 years of age.

6.4 Managed and Natural Stand Definitions

To project stand growth and yield, stands are classified as natural or managed stands based on their year of establishment and leading species. Natural stands are all current stands other than Fir Leading

stands established pre-1973. Natural stand yields are generated using VDYP. Managed stands are considered to be stands all stands except fir leading stands established 1973 and later.

In TSR an analysis was done of existing stands, and fir stands established after 1973 had undergone stand tending and reflected managed stand practices. Stands with other leading species generally had high stem counts and were more represented by a natural yield curve.

Stands that are disturbed through harvesting regenerate to a managed stand, whereas stands that are disturbed by a natural agent and not salvaged regenerate to a natural stand.

6.5 Site Index Assignments

Site index reflects the potential productive capacity of a stand. The inventory site index was used as the site productivity input to develop yield curves for existing natural stands while the managed site index was used for existing managed and future managed stands.

For this analysis, an area-weighted average site index for managed stands was calculated for each leading species type using site indices sourced from the TSR Site Index Adjustment(as described in section 3.5).

6.6 Regeneration

Regeneration assumptions (TISPY inputs) for existing and future managed stands are summarized in Table 22. No stands are assumed to have genetic gains.

These curves will be developed in a multi-part process.

1. Run the planted portion using the inputs, with the total stems per hectare (Natural +Managed), as the SPH for TIPSY.
2. Run the natural portion using the inputs, with the total stems per hectare (Natural +Managed), as the SPH for TIPSY.
3. Create weighted average yield curves using the number of stems of each type.

Table 22 Regeneration Assumptions for Existing and Future Managed Stands

AU Name	Planted Information					Natural Ingress Information		
	Composition	Stems per hectare	Regen Delay (yrs)	OAFs 1	OAFs 2	Composition	Stems per Hectare	Weighted Avg. SI_{50}
1.F-VG	Fdc80 Cw20	1200	1	15	5	Hw80 Cw20	250	33.01
2.F-G	Fdc80 Cw20	1200	1	15	5	Hw80 Cw20	250	27.42
3.F-M	Fdc80 Cw20	1000	1	15	5	Hw80 Cw20	216	22.63
4.F-P	Fdc86 Cw14	1000	1	15	5	Hw50 Fdc30 Cw20	866	17.32
5.CS-GM	Cw50 Sx21 Fdc20 Ba9	600	1	15	5	Cw50 Hw30 Ba20	1200	22.47
6.CS-P	Sx84 Cw7 Yc5 Fdc4	600	1	15	5	Ba40 Sx40 Cw10 Hw10	1913	15.4
7.HB-G	Sx25 Cw23 Fdc20 Ba20 Yc12	350	1	15	5	Fdc30 Ba25 Cw20 Hw20 Yc5	2707	26.37
8.HB-M	Sx25 Cw23 Fdc20 Ba20 Yc12	400	1	15	5	Ba30 Hw25 Fdc25 Cw15 Yc5	1938	22
9.HB-P	Sx25 Cw23 Fdc20 Ba20	600	2	15	5	Fdc30 Cw15 Ba30 Hw20 Yc5	2936	15.99
10. P-A	Fdc40 Cw25 Plc20 Sx15	800	1	15	5	Hw30 Fdc20 Cw20 Ba10 Plc5	1000	18.7

7 Natural Disturbance Assumptions

Disturbances within the THLB are modelled as unsalvaged, or non-recoverable losses (NRL). This analysis adopted the NRL used for the Soo TSA TSR3, which were pro-rated to each tenure's THLB area.

The Soo TSR THLB is approximately 98,000 hectares, and the FNWL THLB is 16,225 hectares, or 16.5% of the Soo. This indicates that the NRLs will be 16.5% of those used in the Soo TSR, which gives an annual NRL of 5,610 m^3/year (Table 23). Harvest flows from the model will be reduced by this amount.

Table 23 Unsalvaged Losses on the THLB

Cause of Loss	TSR	FNWL
Fires	30 000 m^3/yr	4,950
Insects	4000 m^3/yr	660
Total	34 000 m^3/yr	5, 610

8 Modelling Assumptions

8.1 Patchworks™ Model Description

For forecasting and analysis, the PATCHWORKS™ modelling software was used. This suite of tools is sold / maintained by Spatial Planning Systems Inc. of Deep River, Ontario (www.spatial.ca).

Patchworks is a fully spatial forest estate model that can incorporate real world operational considerations into a strategic planning framework. It is unique in its ability to dynamically assess spatial relationships during modelling and adapt solutions to achieve spatial objectives. It utilizes a goal seeking approach and an optimization heuristic to schedule activities across time and space in order to find a solution that best balances the targets/goals defined by the user. Targets can be applied to any aspect of the problem formulation. For example, the solution can be influenced by issues such as mature/old forest retention levels, young seral disturbance levels, patch size distributions, conifer harvest volume, growing stock levels, snag densities, CWD levels, ECA's, specific mill volumes by species, road building/hauling costs, delivered wood costs, net present values, etc. Patchworks continually generates alternative solutions until the user decides a stable solution has been found. Solutions with attributes that fall outside of specified ranges (targets) are penalized and the goal seeking algorithm works to minimize these penalties – resulting in a solution that reflects the user's objectives and priorities.

Patchworks' flexible interactive approach is unique in several respects:

- Patchworks' interface allows for highly interactive analysis of trade-offs between competing sustainability goals.
- Patchworks integrates operational-scale decision-making within a strategic-analysis environment: realistic spatial harvest allocations can be optimized over long-term planning horizons. Patchworks can simultaneously evaluate forest operations and log transportation

problems using a multiple-product to multiple-destination formulation. The model can identify in precise detail how wood will flow to mills over a complex set of road construction and transportation alternatives.

- Allocation decisions can be made considering one or many objectives simultaneously and objectives can be weighted for importance relative to each other. (softer vs. harder constraints)
- Allocation decisions can include choices between stand treatment types (Clearcut vs. partial cut, fertilization, rehabilitation, etc.).
- Unlimited capacity to represent a problem – only solution times limit model size.
- Fully customizable reporting on economic, social, and environmental conditions over time. Reports are built web-ready for easy sharing of analysis results – even comparisons of multiple indicators across multiple scenarios.

8.2 Modelling assumptions

Table 24 Model Assumptions

Criteria	Factor Applied
Blocking	Polygons were grouped into blocks using the built-in patchworks blocking tool (group fragments). Multi-part blocks were created with a target block size of 15 ha. A 20 m distance threshold was used meaning that polygons up to 20 m apart could be considered part of the same block. Blocks were stratified on the following attributes: Analysis Units, and Contributing Classification (i.e. THLB vs NHLB) and were not allowed to contain polygons with more than a 5 year age gap.
Target Block Size	A target block size of 15 ha was used. The blocking tool will attempt to group polygons into 15 ha blocks as long as they meet the specified stratification criteria.
Minimum Block Size	Efforts were made to minimize the incidence of very small blocks (Blocks < 0.1 ha). This is not dependant on the patchworks blocking tool but rather through a GIS eliminate process conducted on the input spatial modelling file.
Planning Horizon	A planning horizon of 200 years will be used split into 10 year periods (20periods x 10 years).

8.3 Harvest Priorities and Target Weightings

The concept of harvest priorities (e.g. oldest first) is not relevant in an optimization/heuristic model. However, within Patchworks, it is necessary to weight various targets or objectives relative to each other so that solutions reflect management priorities. In this analysis, the harvest volume target was weighted substantially lower than all other targets to insure that non timber objectives were not sacrificed to deliver volume. Using this approach harvest volume is attractive to the model only when all other issues have been addressed (e.g. old seral objectives). Weights are set to take into account the scale of different units associated with targets (ha vs m^3 vs %'s).

Patchworks generates millions of alternative solutions and ranks them depending on how well they achieve the user's objectives. For this reason the user must decide when to terminate the search for a better solution. A search is terminated when a specific defined criterion for a 'stable' solution has been achieved. This helps ensure that differences between scenario results occur because of model input differences and not from extra effort spent finding a better solution. For the purpose of this project, Patchwork results were accepted once the objective function improved by less than 0.0001% over 300,000 iterations.

Additionally flow constraints were placed on the model so that for the last 100 years, the growing stock is non-declining, and that after the initial surge in volume, the harvest level remains even.

9 Analysis Report

9.1 Long Run Sustainable Yield Calculation

The Long Run Sustainable Yield (LRSY) is theoretically the largest yield that can be harvested from a forest over an indefinite period. It assumes that stands are harvested when they reach their maximum or culmination mean annual increment (CMAI) and that there is an equal amount of area in each age class by AU (i.e. fully regulated forest). It also assumes there are no non-timber constraints applied. For this analysis, it was calculated by determining the maximum Mean Annual Increment of each future managed AU and multiplying the THLB area within each AU.

Table 25 Long Run Sustainable Yield Calculation.

YG	Effective Area (Ha)	total volume (m^3/yr)	Mean Annual Increment (m^3/ha/yr)
1	1,803	21,286	11.80
2	2,629	21,861	8.31
3	2,217	12,326	5.56
4	1,005	3,737	3.72
5	956	6,736	7.05
6	689	2,418	3.51
7	359	2,916	8.13
8	1,953	11,174	5.72
9	2,859	10,705	3.74
10	57	230	4.04
Total	**14,528**	**93,388**	**5.76**

9.2 Harvest Volume

The harvest assumption for this run is even flow with minimal variation. This harvest must be capable of being maintained with a flat growing stock in the last 50 years.

The base case harvest flow is presented in Figure 9. The volume hovers around 80,000 m^3/ha/yr, with a maximum of 81,606 m^3/ha/yr and a minimum of 78,048 m^3/ha/yr. All harvest volume numbers are net of non-recoverable loss (5, 610m^3/ha).

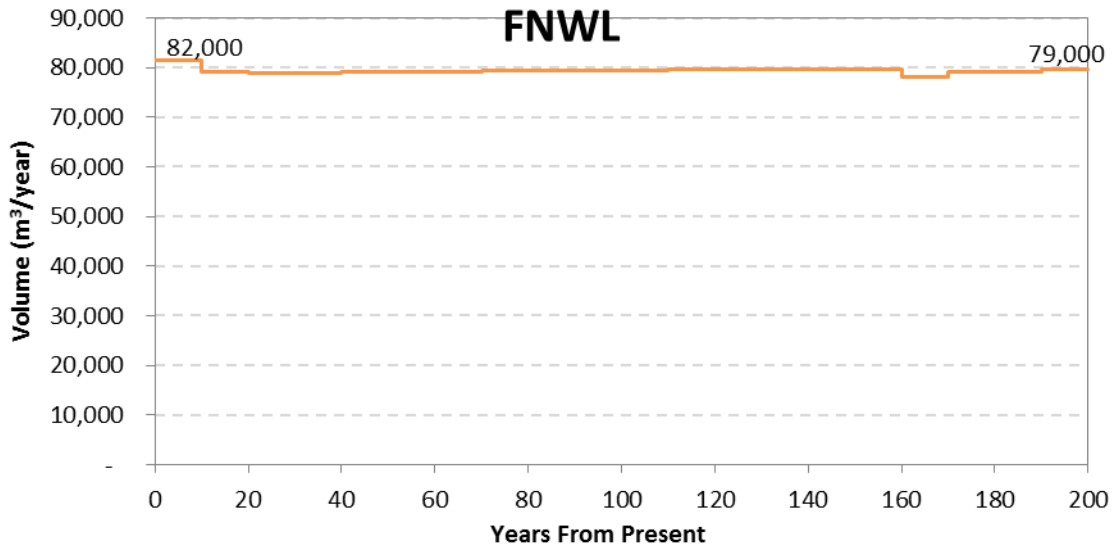

Figure 9 Base case harvest levels

An initial surge of harvest was attempted in the first four periods, but was not available.

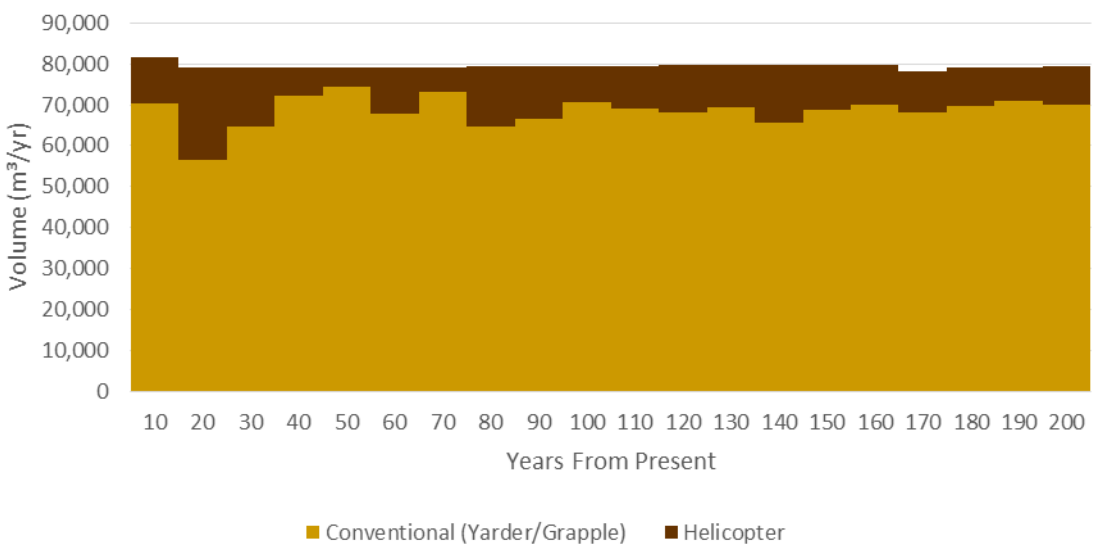

Figure 10 Volume harvested by harvest system.

The landbase was classified for two different logging systems, Conventional (yarder/grapple), and helicopter (Figure 10).The long term harvest flow results in an average of 13% of the volume being harvested from land classified as helicopter access; however, in the short term, this average is much larger at 20% for the first 30 years.

9.3 Growing Stock

The THLB growing stock declines from an initial volume of 3,549,046 m³ to a low of 3,173,348 m³ before period three. It then slowly increases, then dips to keep a steady growing stock over the last 50 years of 3,236,572 m³ (Figure 11). Over time the proportion of Balsam decreases while the volume of Douglas-fir and Cedar increases. This is due to the composition of the managed stands being heavily

cedar and Douglas-fir. This graph also shows the merchantable growing stock over time (starting in the first period). Generally the merchantable growing stock will fluctuate while the age class distribution stabilizes, and over time (longer than this planning horizon) it will even out to a consistent value.

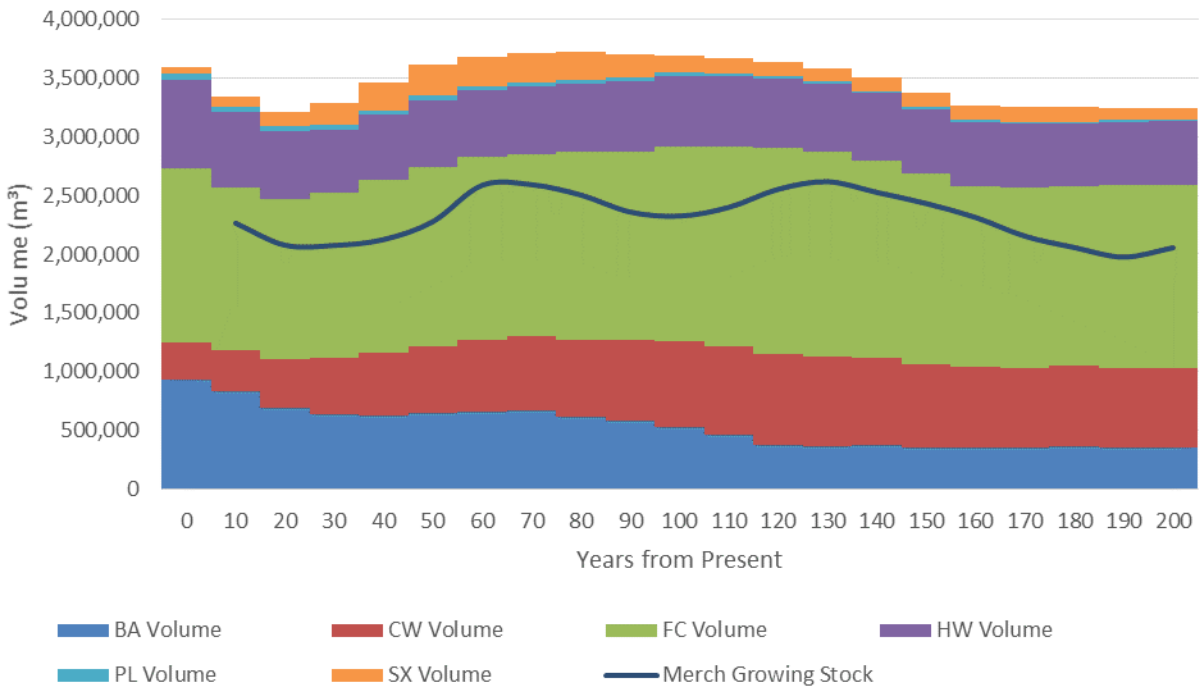

Figure 11 Base Case Growing stock by species.

The timber harvesting landbase is already young. As the landbase converts to managed stands the remaining old in the THLB transitions to younger stands, and the distribution of THLB spans mostly from 0 to 100 years (Figure 12). There is a significant amount of NHLB still present in the 240+ age class distribution.

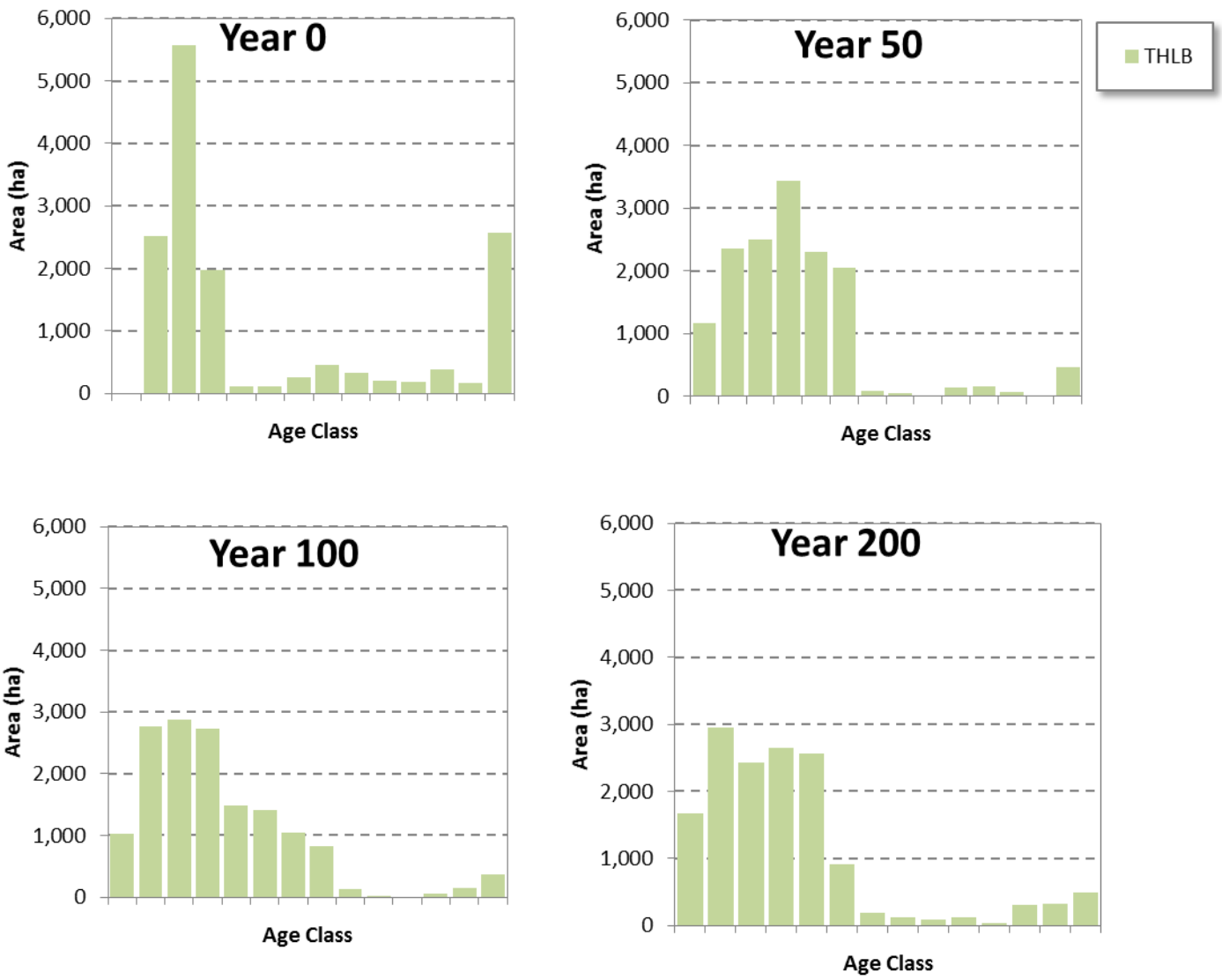

Figure 12 Age class distribution at 0, 50, 100, and 200 years in the future.

9.4 Constraints

The most constraining features on the landbase are the visual landscape inventory Retention and Partial retention polygons. The Modification polygons with the largest THLB contribution are shown in Figure 13[1]. In some periods these polygons come close to the constraints, but they are not continuously constraining.

[1] The percentages in the figures for VQOs are the values for the entire VQO polygon that overlaps the AOI of FNWL, Pemberton CFA, WL1666, and IR 6&7. The VQO's selected here are ordered by largest THLB contribution to the FNWL.

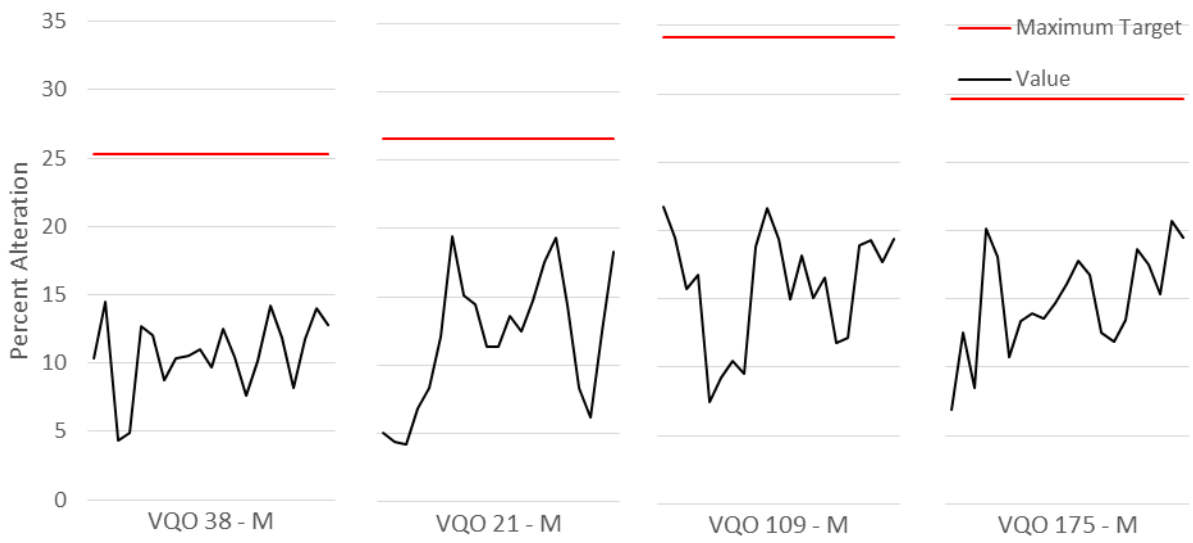

Figure 13 VQO constraints for M - VQO polygons containing the most THLB.

This is not the same for the VQO constraints shown in Figure 14 and Figure 15. These polygons are highly constrained throughout the entire planning horizon.

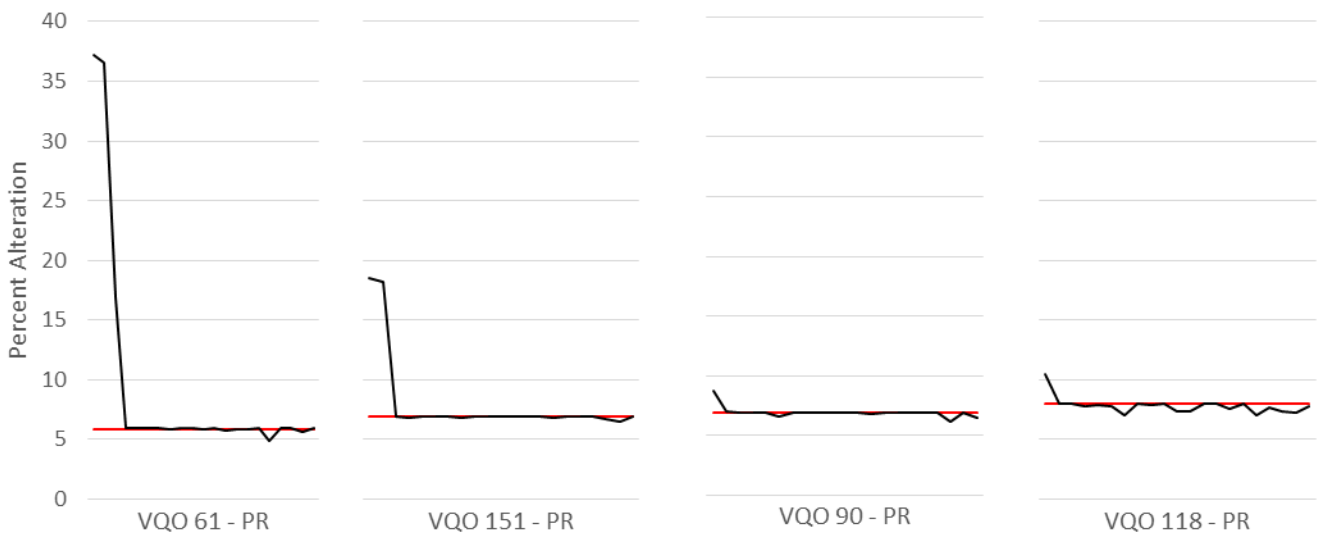

Figure 14 VQO constraints for PR - VQO polygons containing the most THLB

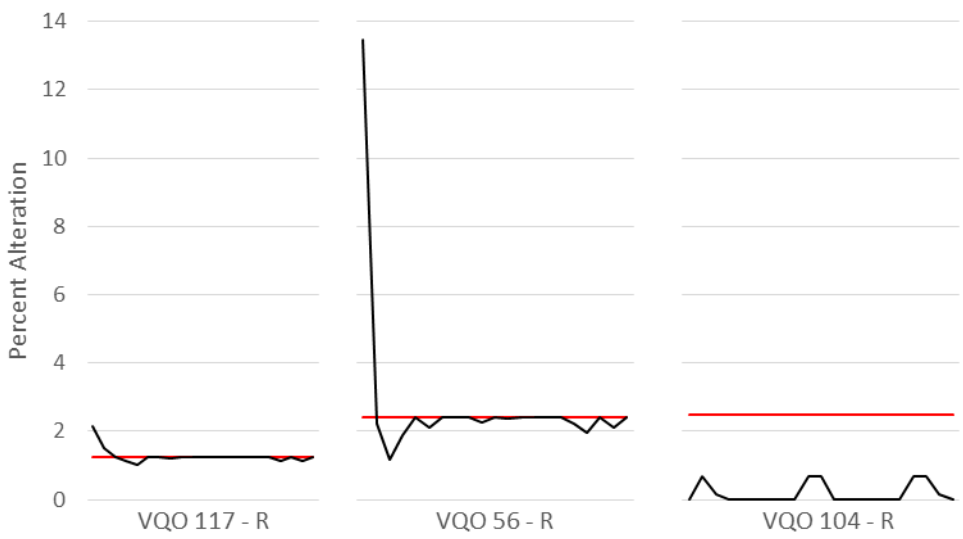

Figure 15 VQO Constraints for R - VQO Polygons (all within FNWL)

10 Discussion

The base case harvest forecast shows a projected cut of 79,380 m³/yr (Averaged across the planning horizon). This value is lower than LRSY because constrains such as the VQOs, the LRMP objectives, and the community watersheds are constraining across the planning horizon. MFLNRO had also projected a hypothetical AAC of 82,249 m³/yr.

Two possible sensitivities that could be calculated are:

1. Inclusion of genetic gain and Fertilization for managed stands
2. Updating heights , Site index, and operability with LiDAR

The first sensitivity would illustrate how investing in stand and stem quality could increase the long term potential of this landbase. The second sensitivity would allow for more accurate calculations on the landbase and give a clearer picture of the capabilities and constraints of this landbase.

Appendix I VLI Polygon Percent and Heights

VLI polygons with no THLB were not included as constraints in the Model.

Table 26 VQO modelling requirements

VLI Polygon Number	Forested Area (ha)	P2P Ratio	Green up Height ht (m)	VQO	ALT	Maximum percent of CFLB under Greenup height (%)
12	107	2.38	6.20	PR	4.3	10.21
13	80	1.98	6.83	M	12.6	24.93
21	1,240	2.11	6.69	M	12.6	26.53
23	468	2.05	6.74	PR	4.3	8.82
33	74	2.21	6.52	M	12.6	27.86
34	28	1.92	6.99	PR	4.3	8.26
35	21	2.92	5.35	PR	4.3	12.55
37	44	1.17	8.37	M	12.6	14.76
38	1,225	2.01	6.81	M	12.6	25.30
44	19	1.32	8.08	PR	4.3	5.67
48	179	1.20	8.26	M	12.6	15.14
51	626	1.38	7.96	PR	4.3	5.93
52	293	1.48	7.81	M	12.6	18.61
53	29	1.49	7.78	M	12.6	18.75
56	684	3.03	5.21	R	0.8	2.42
57	427	1.92	6.98	PR	4.3	8.27
58	1,289	1.64	7.50	PR	4.3	7.06
60	504	1.44	7.87	M	12.6	18.09
61	1,002	1.37	7.98	PR	4.3	5.90
62	661	1.78	7.26	M	12.6	22.45
63	52	1.41	7.91	M	12.6	17.78
71	557	1.45	7.82	PR	4.3	6.26
74	132	1.64	7.49	R	0.8	1.31
75	258	1.78	7.22	PR	4.3	7.66
79	202	1.23	8.24	M	12.6	15.56
80	475	1.28	8.14	PR	4.3	5.49
81	390	2.48	6.06	M	12.6	31.30
82	492	1.73	7.32	M	12.6	21.75
84	2,869	1.77	7.28	PR	4.3	7.60
88	366	1.79	7.21	M	12.6	22.52
89	374	2.02	6.84	PR	4.3	8.66
90	703	1.62	7.58	PR	4.3	6.94
91	1,222	1.63	7.52	PR	4.3	6.99
92	43	1.11	8.43	PR	4.3	4.76
94	392	1.43	7.86	M	12.6	18.02
96	87	1.68	7.44	M	12.6	21.14

97	648	1.46	7.84	PR	4.3	6.26
98	9	1.15	8.34	M	12.6	14.55
99	62	2.20	6.49	PR	4.3	9.44
101	184	2.05	6.80	M	12.6	25.80
104	38	3.10	5.09	R	0.8	2.48
105	657	1.77	7.28	PR	4.3	7.60
106	899	1.98	6.90	M	12.6	24.99
107	174	1.68	7.44	PR	4.3	7.21
109	758	2.71	5.71	M	12.6	34.13
111	154	3.42	4.58	PR	4.3	14.71
113	374	1.94	6.97	PR	4.3	8.35
115	243	1.46	7.82	M	12.6	18.46
117	2,623	1.57	7.62	R	0.8	1.26
118	712	1.87	7.08	PR	4.3	8.03
119	715	2.16	6.57	PR	4.3	9.29
120	361	1.95	6.94	PR	4.3	8.40
122	2	1.32	8.03	R	0.8	1.06
124	289	1.54	7.69	PR	4.3	6.63
128	83	2.95	5.34	PR	4.3	12.68
129	11	1.55	7.69	PR	4.3	6.68
130	446	2.68	5.68	PR	4.3	11.52
131	357	2.83	5.51	PR	4.3	12.16
133	118	2.25	6.38	PR	4.3	9.67
135	137	2.31	6.30	PR	4.3	9.95
136	71	1.66	7.47	R	0.8	1.33
139	180	1.67	7.44	R	0.8	1.33
141	842	1.61	7.57	PR	4.3	6.91
143	92	2.05	6.75	PR	4.3	8.82
144	951	1.62	7.55	PR	4.3	6.96
145	43	2.07	6.72	R	0.8	1.66
148	203	2.72	5.72	PR	4.3	11.71
149	766	1.36	7.99	M	12.6	17.16
150	57	1.71	7.33	PR	4.3	7.34
151	925	1.62	7.53	PR	4.3	6.96
152	449	2.65	5.75	PR	4.3	11.38
157	317	1.50	7.76	PR	4.3	6.43
159	264	1.65	7.49	PR	4.3	7.10
163	173	1.83	7.18	PR	4.3	7.86
166	193	1.52	7.71	M	12.6	19.15
168	1,284	1.75	7.31	PR	4.3	7.52
171	1,043	1.72	7.36	PR	4.3	7.38
175	716	2.35	6.25	M	12.6	29.62
182	569	2.31	6.34	PR	4.3	9.94

186	212	1.70	7.44	M	12.6	21.40
198	14	2.31	6.44	PR	4.3	9.95
204	509	1.79	7.21	PR	4.3	7.72
225	164	2.56	5.90	PR	4.3	11.00
242	0	3.46	4.55	PR	4.3	14.90
605	1,388	1.59	7.58	M	12.6	20.08
606	43	1.63	7.54	M	12.6	20.56
608	371	1.67	7.44	M	12.6	21.07
609	1,465	1.33	8.05	R	0.8	1.07
614	242	2.39	6.25	PR	4.3	10.29
615	87	3.06	5.20	PR	4.3	13.17
616	261	3.19	4.97	PR	4.3	13.70
617	62	3.55	4.43	PR	4.3	15.25